Numerical processing of financial data — Session 6

23/02/2026

Exercise 3

Reminders:

- Black Scholes pricing PDE → Equation (1)
- Call option price (Exercise 1)

$$p^{Call}(t,S) = SN(d_1) - Ke^{-r(T-t)}N(d_2)$$

where $$d_1 = \frac{\log \frac{Se^{r(T-t)}}{K}}{\sigma\sqrt{T-t}} + \frac{1}{2}\sigma\sqrt{T-t}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

- Exercise 3: derive the formula for the put price in the Black-Scholes framework

Exercise 3

1) Show that put-call parity

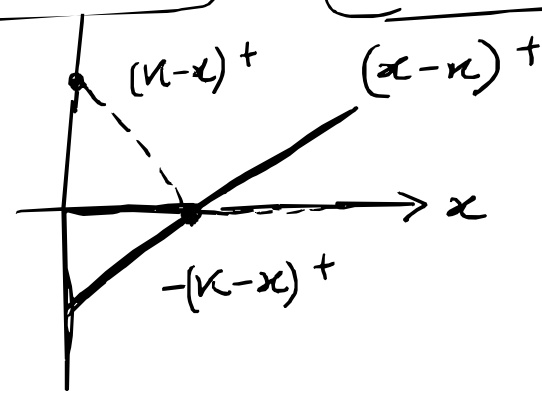
$$\text{Call}_t(T,K) - \text{Put}_t(T,K) = S_t - e^{-r(T-t)}K$$

$$\forall t, \forall T \geq t, \forall K$$

holds in the Black-Scholes framework

Hint:

$$\boxed{(x-K)^+ - (K-x)^+ = x-K} \qquad \boxed{\forall x, \forall K}$$



Hint 2: use the probabilistic representation (from Ex 1)

$$\text{Call}_t^{BS}(T,K) = E\left[e^{-r(T-t)} (Y-K)^+ \right]$$

$$\text{Put}_t^{BS}(T,K) = E\left[e^{-r(T-t)} (K-Y)^+ \right]$$

where $Y = S e^{\sigma\sqrt{T-t}\,G + (r-\frac{\sigma^2}{2})(T-t)}$

then:

$$\text{Call}_t(T,K) - \text{Put}_t(T,K) =$$

$$= e^{-r(T-t)} E\left[(Y-K)^+ - (K-Y)^+ \right]$$

$$= e^{-r(T-t)} E\left[Y - K \right]$$

$$= e^{-r(T-t)} E\left[S e^{(r-\frac{\sigma^2}{2})(T-t) + \sigma\sqrt{T-t}\,G} - K \right]$$

$$= e^{-r(T-t)} e^{r(T-t)} S \, e^{-\frac{\sigma^2}{2}(T-t)} E\left[e^{\sigma\sqrt{T-t}\,G} \right]$$

$$- e^{-r(T-t)} K$$

is this $= 1$?

We use the fact that

$$\boxed{E\left[e^{aG}\right]} = \int_{\mathbb{R}} e^{ax} e^{-\frac{1}{2}x^2} \frac{dx}{\sqrt{2\pi}} = e^{-\frac{1}{2}a^2 \operatorname{Var}(G)}$$

$$\boxed{= e^{-\frac{1}{2}a^2}}$$

$a \in \mathbb{R}$, $G \sim N(0,1)$

↳ Laplace Transform of a Gaussian random variable

Hence, we get:

$$e^{-\frac{1}{2}\sigma^2(T-t)} E\left[e^{\sigma\sqrt{T-t}\,G} \right] = \cancel{e^{-\frac{1}{2}\sigma^2(T-t)}} \, \cancel{e^{\frac{1}{2}\sigma^2(T-t)}} = 1$$

▷ Therefore, we conclude that the relationship

$$\boxed{\text{Call}_t(T,K) - \text{Put}_t(T,K) = S - e^{-r(T-t)} K}$$

holds

Note: the functions on the LHS depend on σ (the volatility parameter in the BS formula), but the RHS does not.

2) From call - put parity:

$$P_t(T,K) = Call_t(T,K) - S + e^{-r(T-t)}K$$

$$= SN(d_1) - Ke^{-r(T-t)}N(d_2) - S + e^{-r(T-t)}K$$

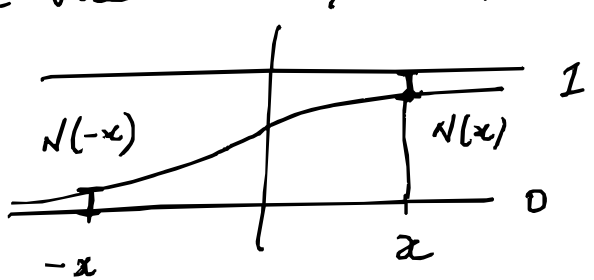
from Ex 1

$$= -S\underbrace{(1 - N(d_1))} + Ke^{-r(T-t)}\underbrace{(1 - N(d_2))}$$

- We use the property:

$$1 - N(x) = N(-x) \qquad \forall x \in \mathbb{R}$$

still true to the symmetry of the gaussian distribution.



- We get

$$\boxed{P^{put}(t,S) = P_t^{BS}(T,K) = -S\,N(-d_1) + Ke^{-r(T-t)}N(-d_2)}$$

where d_1 and d_2 are the same coefficients

3) <u>Easy consequence of call-put parity</u>:

We already know (from Ex 2)

$$\text{Call delta} = \partial_S p^{Call}(t,S) = N(d_1)$$

$$\text{Call Vega} = \partial_\sigma p^{Call}(t,S) = S\sqrt{T-t}\, n(d_1)$$

$$\text{where } n(x) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}$$

Now, from parity:

$$p^{Put} = p^{Call} - S + K e^{-r(T-t)}$$

then:

- $$\boxed{\partial_S p^{Put}(t,S)} = \partial_S p^{Call}(t,S) - 1$$
$$= N(d_1) - 1 = \boxed{-N(-d_1)}$$

- $$\boxed{\partial_\sigma p^{Put}(t,S)} = \partial_\sigma p^{Call}(t,S) + 0$$
$$= \boxed{S\sqrt{T-t}\, n(d_1)}$$

$\Rightarrow$ Vega of the put = Vega of the call option

<u>Note:</u> for the Gammas, we also have:

$$\boxed{\partial_{SS} p^{Put}(t,S) = \partial_{SS} p^{Call}(t,S)} \rightarrow \text{same Gammas}$$

again from put-call parity.

<u>Comment</u>: call-put parity is related to the fact that

$$\text{Call} - \text{Put} = \text{linear contract on } S$$

$$(S_T - K)^+ - (K - S_T)^+ = \underbrace{S_T - K}_{\text{linear contract on } S} \qquad \forall K,$$

for every value of S_T

$\Longrightarrow$ <u>put-call parity</u> is <u>always true</u> (also outside the BS framework) if there is no arbitrage on the market.